

Delta Lake MERGE Implementation Report

Objective

Perform incremental data processing using Delta Lake by loading, cleaning, merging, validating, and displaying data.

Introduction

Delta Lake provides ACID transactions, schema enforcement, time travel, and MERGE support on Apache Spark.

Technologies

Python 3.11
Apache Spark 3.5.6
Delta Lake 3.2.0
Java 17
VS Code

Methodology

Loaded dataset, cleaned data, renamed columns, created Delta table, created incremental dataset, performed MERGE, validated results, displayed final dataset.

Results

Original Rows: 9994
Final Rows: 9996
Updated one existing record
Inserted two new records
No duplicate Row_ID values.

Conclusion

The assignment successfully demonstrates incremental processing using Delta Lake MERGE.